

27th-09-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

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Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Title: INTRODUCTION TO PROGRAMMABLE LOGIC
CONTROLLERS

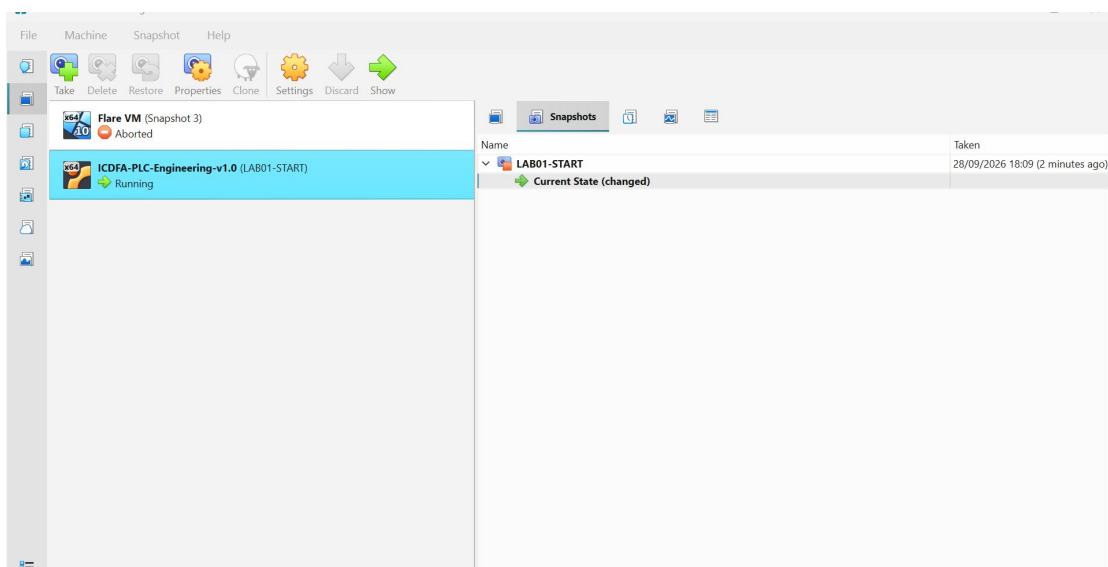
Assignment: Week 1-Lab 1 Report
C11/26/ACIS/17359

Instructor : Aminu Idris

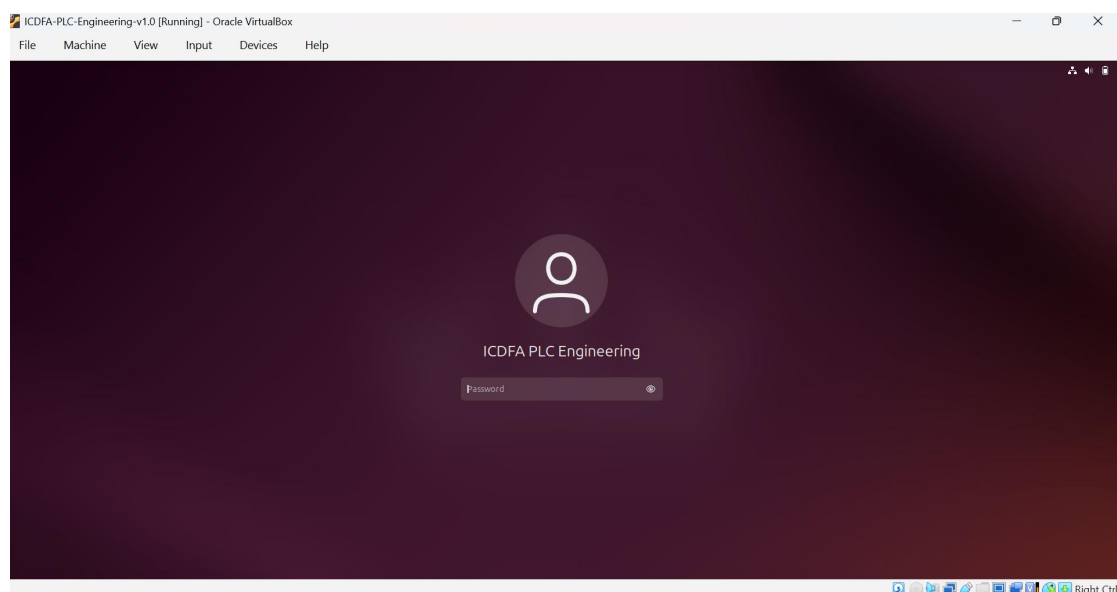
Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	27 th -09-2026

Steps 1-3: Setup

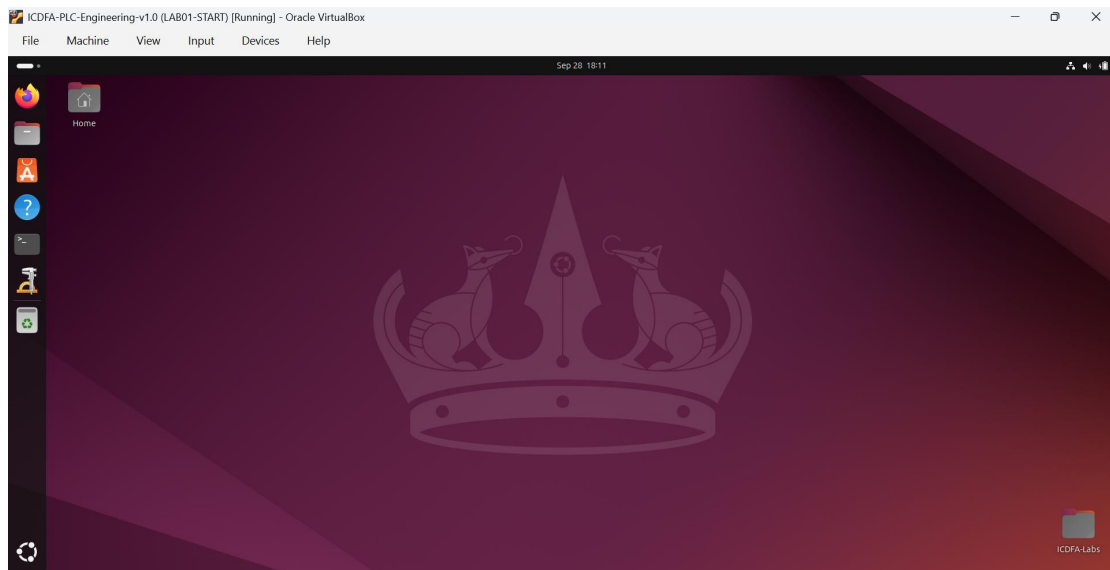
1. Import the VM into VirtualBox (File > Import Appliance).
2. Set Network > Adapter 1 to **Host-only Adapter** (or Internal Network). Do not use Bridged.



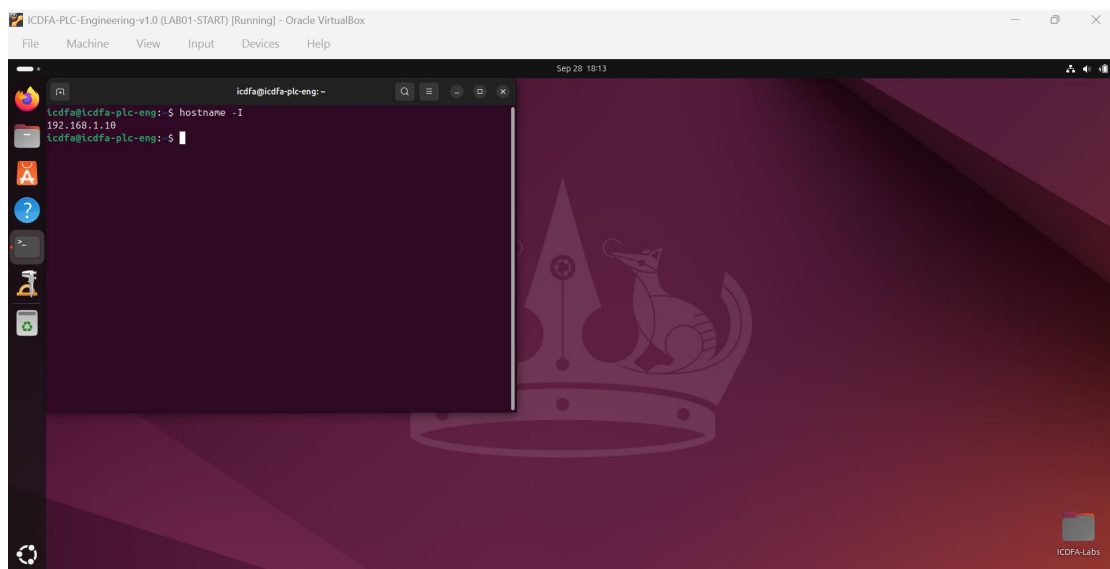
3. Take a snapshot named LAB01-START.



4. Start the VM and log in with plc / password.



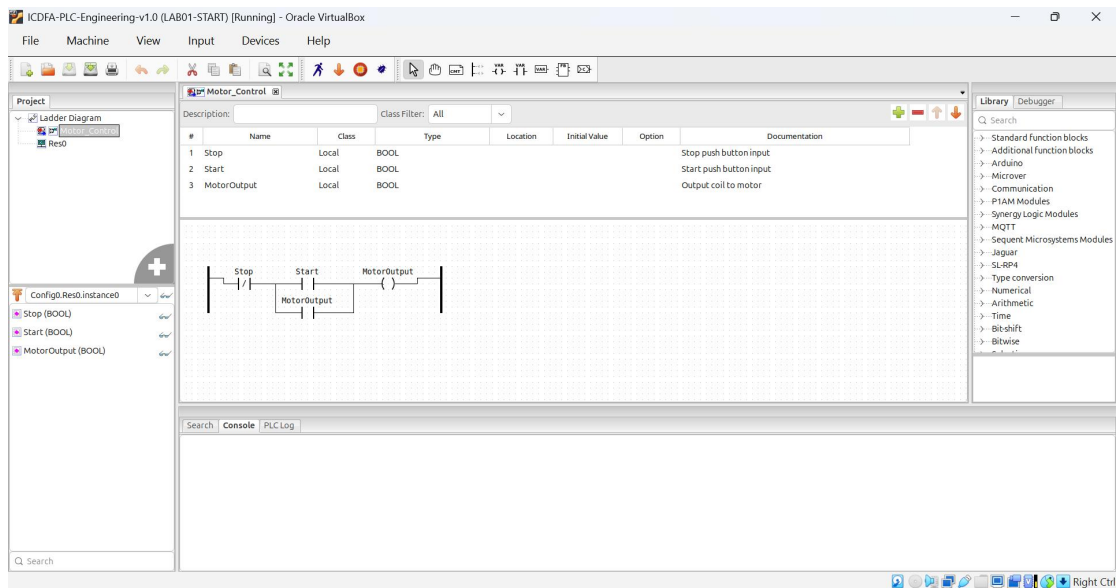
5. Run `ip a` and confirm the address is `192.168.1.10`. If it differs, record the value in the report and don't change it.



ip confirmed 192. 168. 1. 10

Steps 4-6: Open the program

1. Open **OpenPLC Editor** from the desktop or application menu.
2. Go to **File > Open** and browse to *PLC Programming Language Examples > Ladder Diagram*.
3. In the Project tree, expand **Res0**, then click the edit icon beside **instance0 (Motor_Control)**.
4. Use the fullscreen toggle and drag the pane wider so the whole rung is visible.
5. Take your screenshot now, since it is required evidence.



Step 7: Identify the elements

Element	Symbol	Role
Left and right power rails	Vertical lines	Left rail is the power source, right rail is the return. Logic flows left to right.
Start	Normally open (NO) contact	Input. Closes only while the Start button is pressed.
Stop	Normally closed (NC) contact /	Input. Passes power until the Stop button is pressed.
MotorOutput contact	NO contact in parallel with Start	The latch (seal-in). It closes when the motor is on, so it keeps the rung energized after Start is released.
MotorOutput coil	()	Output. Energizes the motor.

Reflection

Network-connected PLCs let operators monitor and control processes remotely, which brings real operational benefits. Engineers can change setpoints, collect data for maintenance and efficiency, and integrate machines from different vendors without rewiring relay panels. This reduces downtime and supports central supervision of widely distributed assets such as pipelines and water plants.

However, connecting PLCs to networks also expands the attack surface. Many industrial protocols, such as Modbus/TCP, were designed for reliability rather than security and often lack authentication or encryption. A device that was once physically isolated may now be reachable from corporate IT networks or the internet. Default or weak credentials, unpatched firmware, and poor segmentation make this

worse. Because PLCs control physical processes, a compromise can affect safety, equipment, and public services rather than just data.

Organizations therefore need to balance connectivity with protection. Sensible measures include network segmentation, isolating control networks from business networks, changing default credentials, restricting remote access, monitoring traffic, and keeping tested backups of PLC programs. Availability and safety are typically the priority in operational technology, so security controls must be introduced carefully so they don't disrupt the process.

Knowledge check

1. Input module vs output module?

An input module receives signals from field devices (buttons, switches, sensors) and passes them to the processor. An output module sends the processor's commands to load devices (motors, solenoids, pilot lights).

2. Why a normally closed contact for stop/emergency-stop?

An NC contact conducts by default and opens when pressed. If the wire breaks or the device fails, the circuit opens and the equipment stops, so a fault leads to a safe state (fail-safe design).

3. Discrete and analog examples

Discrete: a push button or limit switch (ON/OFF).

Analog: a temperature or tank-level sensor giving a continuous range (for example 4-20 mA).

4. Operational risk of network-connected devices

Networked devices can be reached by unauthorized or malicious traffic.

Unauthorized commands or altered logic can stop production, damage equipment, or create unsafe conditions, and a fault on the network can disrupt control of the process.